Six clues hidden in water

H2O: Why Water Is Special

Name _____ Date _____



Little held a clear glass to the light. Tiny molecules moved inside. Every water molecule has the same recipe: two hydrogen atoms joined to one oxygen atom. H2O records that recipe.



An ice cube softened into a puddle. Solid ice had melted into liquid water. Water can also become a gas called water vapor. These changes rearrange the molecules, but they remain water molecules.



Little placed drops on wax paper. Each curved into a bead instead of spreading flat. Water molecules attract one another, pulling the surface together. Two nearby beads joined into one larger drop.

water	a substance made of H2O molecules
molecule	atoms joined as one tiny unit
atom	a tiny piece of matter
hydrogen / oxygen	the H and O atoms in H2O
ice	solid water
melt / freeze	solid to liquid / liquid to solid
vapor	water in its gas state
surface tension	a liquid surface resisting a pull
float	stay at a liquid's top
dissolve	spread through a liquid in tiny pieces

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Big lowered a paperclip on a small raft of tissue. The tissue sank, but the clip stayed at the top. The water was not covered by a solid skin. Its surface tension came from water molecules pulling together, making the surface resist being broken.



In liquid water, molecules slide close together. When water freezes, the molecules lock into a roomier pattern. That makes ice lighter for its size than liquid water, so it floats. On a pond, floating ice forms first at the top while liquid water—and fish—can remain below.



A spoonful of sugar seemed to disappear in water, but its tiny pieces had not vanished. Water molecules surrounded those pieces and carried them through the glass. Water can dissolve many substances, including sugar and salt, though it cannot dissolve everything. One small molecule had many remarkable ways to act.

1. What does H2O tell you about the atoms in one water molecule?
2. How do both rounded drops and a resting paperclip give evidence that water molecules attract one another?
3. Why can floating ice help fish survive beneath a frozen pond?

Your turn

Write one clue that shows why water is special.

Example: Ice floats because it is lighter for its size than liquid water.